

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Adaptation, terminology, succession & population control - Set 1

Mark scheme

AQA spec 3.1.3; The living environment; textbook pp. 72-85

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• The range of an environmental factor within which a species / population can survive (survival highest at the optimum, falling towards the limits) • X survives from about pH 5.5 to 8.5 (wide range); Y only from about pH 6.0 to 8.0 (narrow range) • Both have the same optimum (pH 7.0)	3
01.2	• X: 60% survive at pH 6.0 whereas only 5% of Y survive • X has a larger gene pool / wider range of tolerance so some individuals are adapted to the new conditions • Survivors breed and their offspring are adapted so the population rebuilds; Y is likely to die out locally	3
01.3	• Reveals the habitat requirements / which abiotic factor is near the edge of tolerance • Management can make small beneficial changes to that factor (eg liming, shading, water levels) to keep conditions within the range and increase survival	2
01.4	<i>Any 2 from:</i> • Food supply • Pollinators • Seed dispersal species • Disease / pathogens • Predators / competitors	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-73; AQA spec 3.1.3.1.	

Question	Marking guidance	Marks
02.1	• Species: a group of closely related organisms that naturally interbreed to produce fertile offspring and are reproductively isolated from other groups (Passer domesticus, house sparrow); the only 'real' taxon • Niche: the role a species plays in its habitat - how it uses resources and relates to other species (tawny owl: nocturnal carnivore nesting in tree holes eating small mammals); two species cannot share exactly the same niche in one habitat • Community: all the populations of all species (plants, animals, fungi, bacteria, protists, archaea) living in an area • Biome: a large geographical region with a specific climate and characteristic community, including all areas where it occurs (all tropical rainforests, tundra, savannah)	8
02.2	• An ecosystem is the biotic community plus abiotic factors of one area, relatively self-contained (a particular coral reef) • A biome includes all separate ecosystems with the same climate and community type (all coral reefs); each reef is a different ecosystem within the same biome	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 82-85; AQA spec 3.1.3.2.	
03.1	• Hydrosere • Single-celled algae (and floating plants) - arrive on birds / insects / from soil	2
03.2	• Pioneers and later plants grow and die; dead organic matter plus sediment washed in fill the pond so water becomes shallower • As depth falls rooted plants (pondweeds then emergent reeds with leaves above water) can anchor and colonise; emergents shade submerged plants which die out • Sediment / soil accumulates and becomes able to support larger plants; water-tolerant trees (willow, alder) colonise the waterlogged soil • Transpiration by trees and further sediment dry the soil and make it aerobic so oak can colonise; the denser canopy shades out smaller plants • Each community changes abiotic conditions (depth, light, nutrients, oxygen), making it suitable for the next colonisers which out-compete them; conditions become less extreme and biotic factors increasingly important	5
03.3	• Starting conditions (bare rock vs open water) differ but each stage makes conditions less extreme and builds soil / organic matter • The final community is controlled by the regional climate (temperature, water availability), which is the same for both - climatic climax (eg oak woodland in the UK)	2
03.4	• Lichens and algae colonise bare rock (extreme temperatures, no soil, little water); rock fragments and dead organic matter accumulate; mosses, then grasses and ferns as thin soil forms; wind-dispersed trees (birch) then animal-dispersed (oak, beech)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 73-76; AQA spec 3.1.3.3.	

Question	Marking guidance	Marks
04.1	• Primary: succession on a newly created habitat with no life / no soil (bare rock, new pond, sand) • Secondary: succession restarting after an existing community is disturbed (tree felling, ploughing, fire, end of grazing) • Faster because soil already exists and many seeds / roots / organisms are still present so there are fewer delays for colonisation	3
04.2	• Human activity (grazing, mowing, burning, coppicing) regularly removes the later successional species so the climatic climax cannot form • The stable community that develops under this repeated disturbance is a plagioclimax • Examples: chalk grassland / hay meadows maintained by grazing or mowing; lowland heath by burning; coppiced woodland; Machair grassland; Broads grazing marshes	3
04.3	• If the traditional activity stops, secondary succession returns the habitat to scrub / woodland and the plagioclimax species are lost • Rainforest and coral reefs are climax communities so conservation aims to minimise human impact	2
04.4	• Extreme environments (deserts, polar) are dominated by abiotic factors; few species tolerate them; populations fluctuate and food webs are simple so changes in one food species have large effects • Stable environments allow many species to evolve inter-species relationships; populations are regulated by biotic factors; complex food webs give stability (rainforests, reefs)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 76-77; AQA spec 3.1.3.3.	
05.1	• Sigmoidal (S-shaped) growth curve • Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate • Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate • Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate	4
05.2	• Density-dependent: effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation • Density-independent: chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter	2
05.3	• Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density • At this density deer over-graze, preventing tree regeneration and damaging habitats / other species; culling keeps numbers below the damaging level (as in Scotland)	2
05.4	• Rising prey numbers give predators more food so predator numbers rise; high predation then reduces the prey population • Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.	

Question	Marking guidance	Marks
06.1	• Anchovy r-selected; blue whale k-selected • r-selected: mature quickly, produce many young, disperse widely, short lives; adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) • k-selected: mature late, few young with parental care, long-lived; populations stable near carrying capacity (whales, elephants, rhinos)	3
06.2	• Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly • An increase in death rate above the low birth rate causes a population crash • Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium); r-selected species rebound within a few years once exploitation stops	3
06.3	• New population = starting population + births + immigrants - deaths - emigrants	1
06.4	• The greatest rate of exploitation (harvest) possible without causing long-term population decline • Equals the surplus production: highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) • At carrying capacity net growth is zero so any harvest reduces the population; harvesting above MSY causes decline	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Succession: pioneers, seres (hydrosere, lithosere, psammosere), modification of abiotic conditions, climax communities controlled by climate; deflected succession and plagioclimaxes • Application: deciding whether to manage (plagioclimax habitats need grazing, mowing, burning, coppicing, pollarding continued) or leave alone (climax habitats); habitat creation planning the stages; controlling scrub / secondary succession; understanding that new habitats change over time • Population dynamics: birth and death rates, density-dependent and independent factors, carrying capacity, sigmoid growth, predator-prey cycles, r- and k-selection, MSY • Application: setting sustainable harvest levels (fisheries, whaling, forestry); recognising k-selected species need strict protection and recover slowly; deciding when culling is needed (deer without wolves, introduced species) or when releases / captive breeding are needed (low breeding rate); predicting effects of reintroducing predators • Monitoring population size, distribution, survival rate and age structure to assess success; adaptation and range of tolerance guiding abiotic management • Evaluation: uncertainties (natural fluctuations, unknown carrying capacity); examples (Yellowstone wolves, Scottish deer, chalk grassland); judgement	9
07.2	• Any taxonomic group of organisms classified together by similarity (eg species, genus, family)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 72-85; AQA spec 3.1.3.	